

IngeniumAI Hackathon – Non-contrast aorta & PA segmentation
Compiled 2026-08-13 15:08 UTC

Task: segment pulmonary artery and aorta on non-contrast chest CT
so PHAST can run earlier than contrast CTPA.

Method: contrast-removal domain adaptation (not plain supervised).
Source = unused contrast CTPA. Vessel HU is flattened to soft-tissue
range so SegResNet must use shape and location. Mixed with held-in
non-contrast cases under the same HU randomization.
Val = 3 held-out non-contrast cases (never trained on).

Device: mps

Non-contrast paired: 10 Contrast paired: 13

Mean hold-out Dice (non-contrast val)

Baseline	PA 0.000	aorta 0.068	mean 0.034
Adapted	PA 0.000	aorta 0.349	mean 0.175

Clinical mean equivalent diameter at max-PA GT slice

GT	MPA 56.9 mm	aorta 42.6 mm	ratio 1.317515847443455
Base	MPA 0.0 mm	aorta 13.9 mm	ratio None
Adapt	MPA 0.0 mm	aorta 57.7 mm	ratio None

Licence: CT-RATE is CC BY-NC-SA 4.0 (demonstration only).

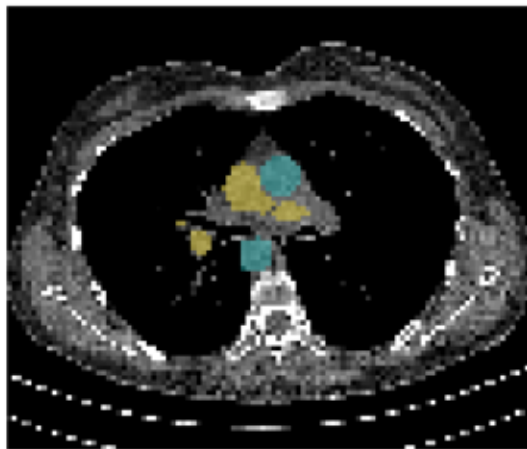
A production PHAST model would be retrained on licensed/clinical data.

FUMPE / CAD-PE contrast sets are CC BY 4.0.

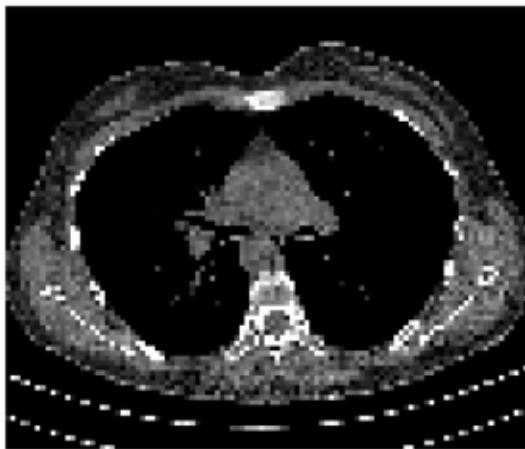
baseline train 10000 a 2.nii

baseline train_10000_a_2.nii PA Dice 0.00 aorta 0.00

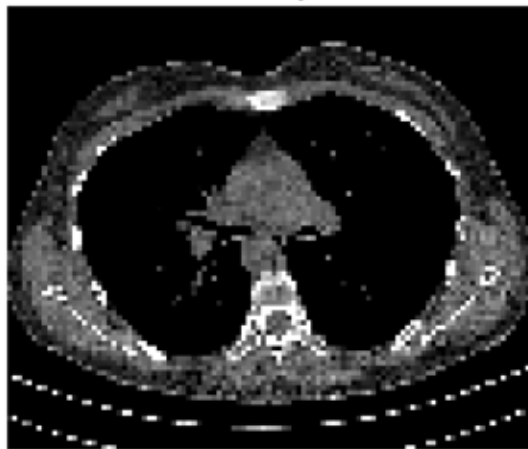
Ground truth



Prediction



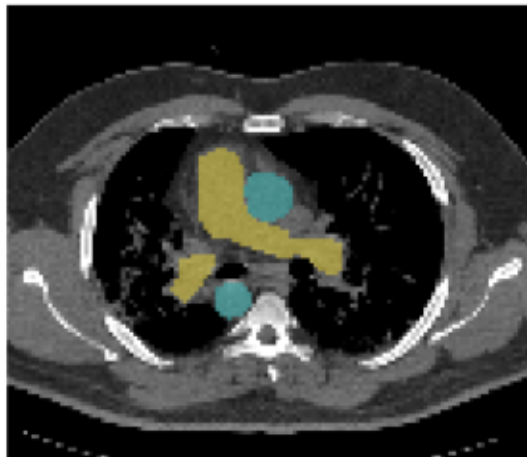
CT only



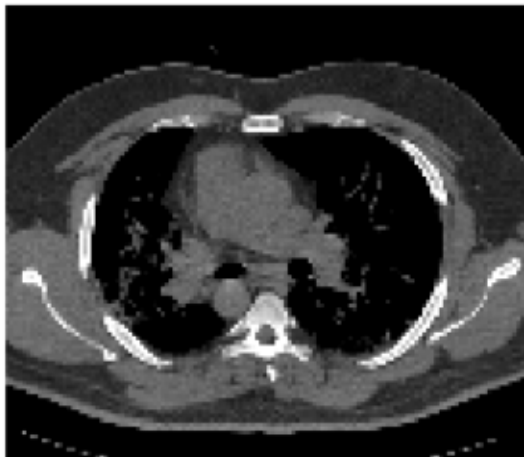
baseline train 10019 a 2.nii

baseline train_10019_a_2.nii PA Dice 0.00 aorta 0.02

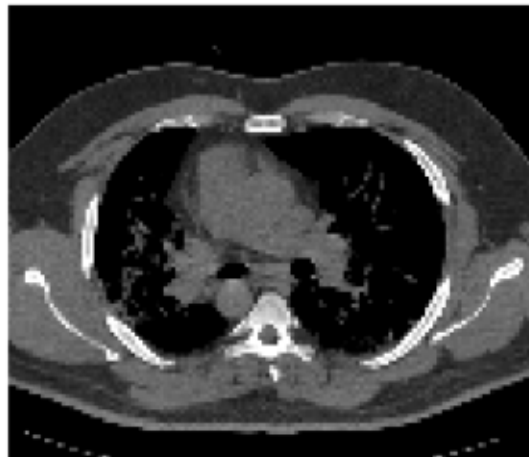
Ground truth



Prediction



CT only



baseline train 10020 a 2.nii

baseline train_10020_a_2.nii PA Dice 0.00 aorta 0.19

Ground truth



Prediction



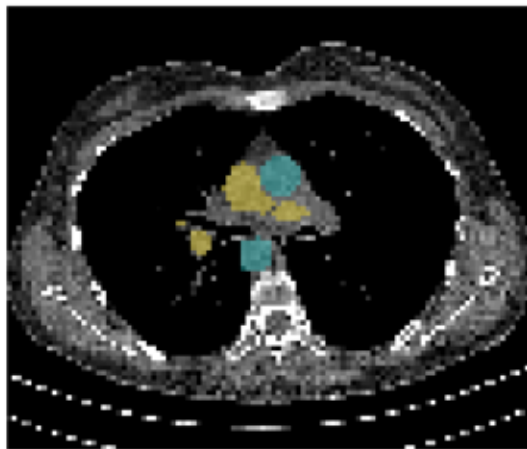
CT only



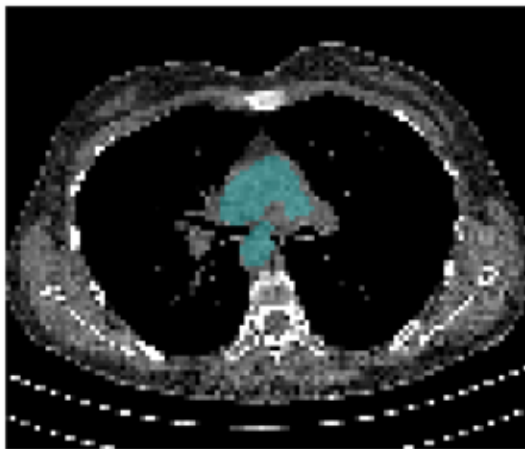
adapted train 10000 a 2.nii

adapted_train_10000_a_2.nii PA Dice 0.00 aorta 0.52

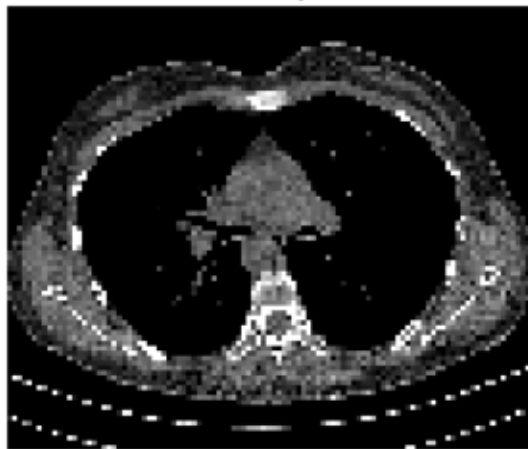
Ground truth



Prediction



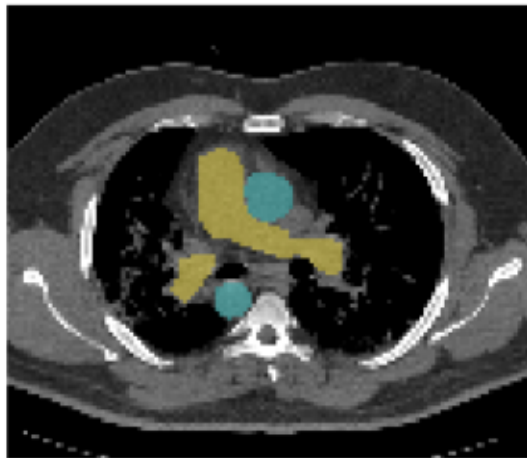
CT only



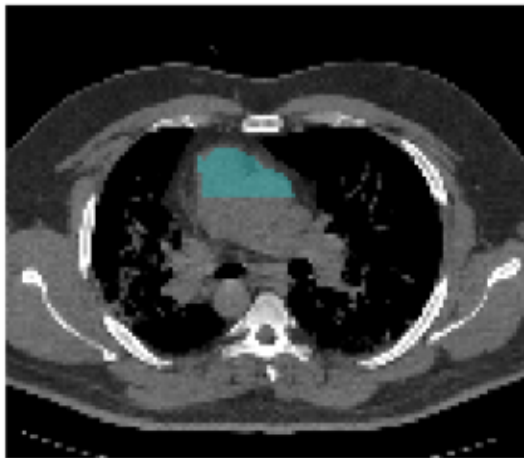
adapted train 10019 a 2.nii

adapted_train_10019_a_2.nii PA Dice 0.00 aorta 0.08

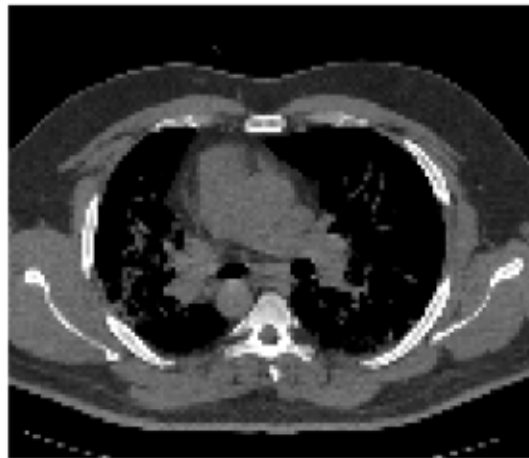
Ground truth



Prediction



CT only



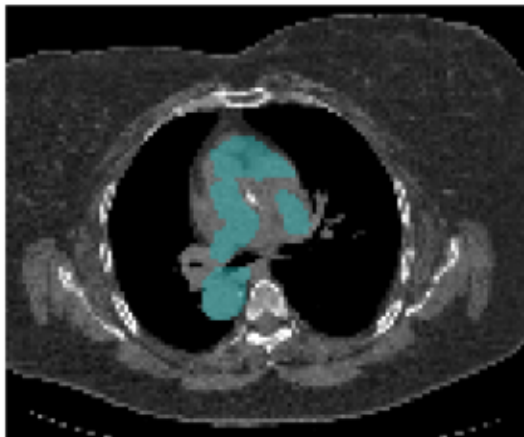
adapted train 10020 a 2.nii

adapted_train_10020_a_2.nii PA Dice 0.00 aorta 0.45

Ground truth



Prediction



CT only

